

Alexander He

Curriculum Vitae

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EDUCATION

MSc in Statistics

University of Toronto

Sept 2026 – Apr 2027

- Advisor: Prof. Ting-Kam Leonard Wong

HBSc in Statistics and Economics

University of Toronto - St. George Campus

Sept 2022 – Apr 2026

- Economics and Mathematics Specialist (ASSPE2599), Specialist in Statistical Sciences; Theory and Methods (ASSPE2290)

SELECT COURSEWORK

- **Real Analysis (PhD-level):** Carathéodory Extension, differentiation and integration of measures, Vitali Covering Theorem, Lebesgue Density Theorem, Riesz-Markov-Kakutani Representation, Radon-Nikodym
- **Probability Theory (PhD-level):** Measure theory, expectation and integration, inequalities and modes of convergence, stochastic processes, Fubini's Theorem, weak convergence
- **Optimal Transport (MSc Reading Project):** Monge and Kantorovich formulations, c -cyclical monotonicity, Wasserstein space, dual problems, entropic regularization, statistical convergence, generalized Wasserstein distances and transport problems, applications to topology
- **Statistical Genetics (Graduate):** Population genetics, GWAS, Hardy-Weinberg Equilibrium, pedigree likelihood, parametric and non-parametric linkage, multiple hypothesis testing
- **Multivariate Statistics:** Multivariate Normal distribution, Principal Component Analysis, UMAP, Canonical Correlation Analysis, Independent Component Analysis, multivariate ANOVA, Factor Analysis, kernel methods

RESEARCH EXPERIENCE

University of British Columbia Department of Electrical and Computer Engineering

Post-Undergraduate Researcher, Deep Medical Imaging and Optimal Transport

June 2026 -

- Developing a modality-robust representative sample-selection algorithm for 3D glioma image segmentation (UPenn BraTS Challenge data) via multimodal VAEs (PyTorch, NiBabel and MultiVae Python libraries)
- Greedy selection approach via Bures-Wasserstein distance of Gaussian distributions from MVAE representation for small annotation budgets
- Performance metrics via SwinUNETR base model against active learning approaches (Random sampling, Monte Carlo dropout) using the MONAI library
- Supervisors: Prof. Zhen Jane Wang, Xiaohui Chen (University of Southern California)

University of Toronto Data Sciences Institute

Summer Undergraduate Data Sciences (SUDS) Internship Program

May 2025 - Feb 2026

- Research assistant in data science for AI text classification task for early childhood education documents, analyzing disability and inclusion related content via LLMs (OpenAI API in Python, langchain-openai library)
- Wrote Python scripts for parsing CSVs from LLM output, OOP, optimizing LLM pipelines, hallucination mitigation; average performance metrics around 80% on pilot data
- Version control via GitHub (Private repository)
- Supervisor: Prof. Elizabeth Dhuey

University of Toronto Research Opportunity Program

Research Opportunities Course: ESS299, Research Assistant

Apr 2024 - Mar 2025

- Statistics and geoscience research position under the Department of Earth Sciences, using Convolutional Neural Networks (With the graduate team) and government legacy data to predict gold deposits in the Yukon region
- Finalists for Prospector's and Developer's Association of Canada (PDAC) Frank Arnott NGEA Challenge 2025
- Studying machine learning techniques including Principal Component Analysis and clustering in R
- Geospatial analysis in OpenStreetMap, QGIS, exploratory data analysis via ioGAS
- Supervisors: Prof. Daniel Gregory and Andrei Swidinsky

TEACHING

Teaching Assistant

STA130: An Introduction to Statistical Reasoning and Data Science

Jan 2027 -

- Lead weekly tutorials, grade coursework, and host office hours for students

AWARDS

SUDS Opportunities Program Award

University of Toronto Data Sciences Institute

May - Aug 2025

- \$7,200 funding award from the Data Sciences Institute as part of the SUDS cohort

Next Generations Explorers Award 2025

Prospectors and Developers Association of Canada

Mar 2025

- Impact & Exploration Prize Winner, Runner-up award of \$3,000 at PDAC's NGEA conference

Dean's List Scholar x3

University of Toronto Woodsworth College

2024, 2025, 2026

- Awarded for outstanding academic promise and achievement for the 10th, 15th and 20th credit obtained at UofT (Summer 2024, Winter 2025, Winter 2026)

VOLUNTEERING EXPERIENCE

Kensington Gardens Volunteer - Kensington Health

Friendly Visitor

May 2025 - Mar 2026

- Providing weekly companionship and visiting to hospice residents, assisted in reporting resident status and safety
- Completed rigorous training modules on resident rights, volunteer responsibilities, ethics, grief and bereavement, workplace health and safety, psychosocial support and more

OTHER

- **Programming:** Python, R, Stata
- **Python Libraries:** PyTorch, MONAI, LangChain OpenAI, MultiVae, GeomLoss, SimpleITK, NiBabel, POT, PyKeOps, NumPy, pandas, Seaborn, Matplotlib, scikit-learn, RapidFuzz, NLTK
- **R Packages:** tidyverse, caret, transport, sandwich, ivreg, survey, MASS, lme4
- **Software:** GitHub, LaTeX, QGIS, ioGAS, Geoscience ANALYST, Excel, PLINK 2
- **Citizenship:** Canadian
- **Languages:** English (native), Mandarin (fluent), French (basic)